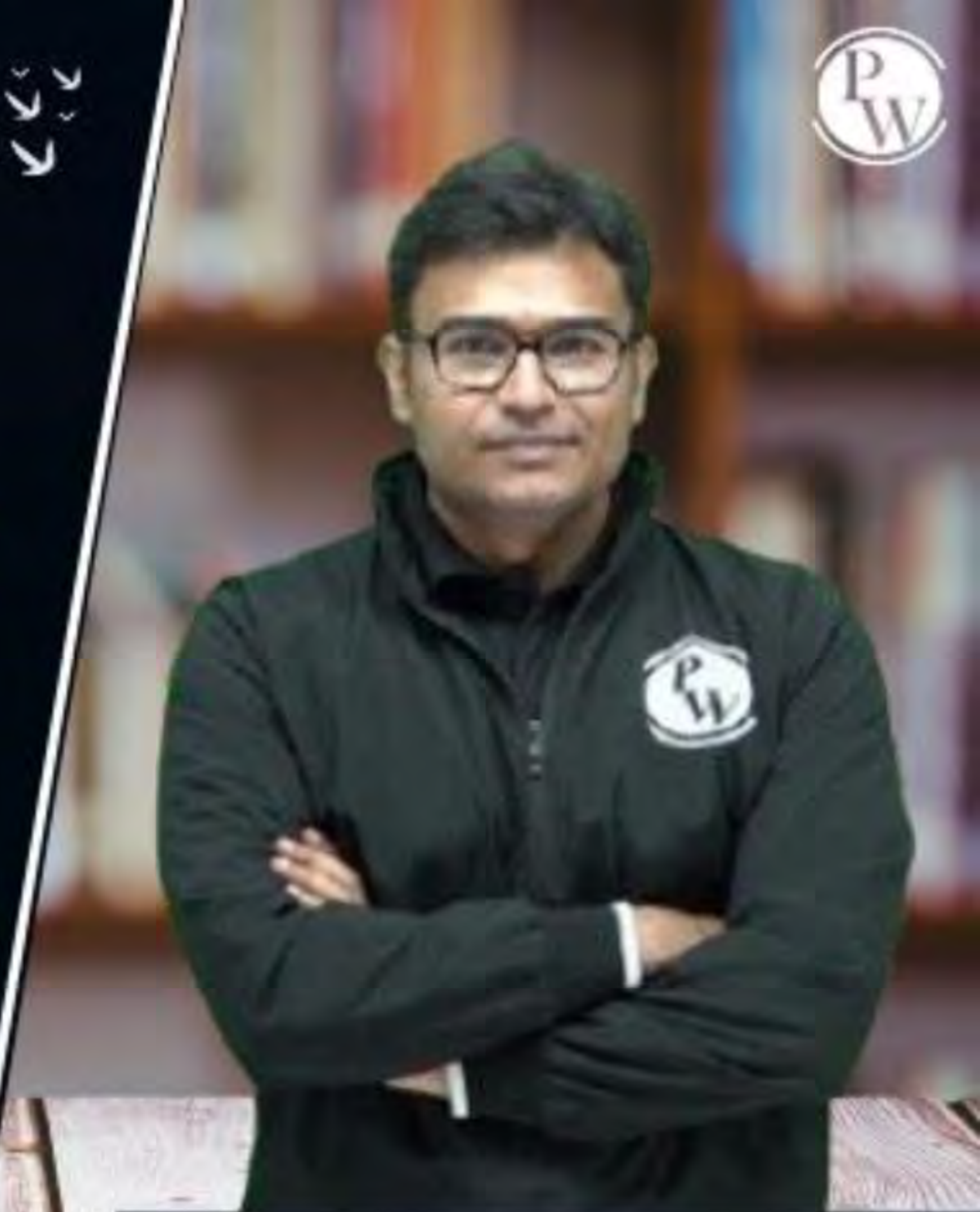
Computer Science & IT

Theory of Computation

Regular Languages

Lecture No. 8



By- Ravindra Sir

Recap of Previous Lecture

Topic DFA, NFA, RE

Topic

Topic

Topics to be Covered

Topic

Grammars

Topic

Topic

$L = \{a\}$

$S \to a$

$L = \{a, b\}$

$S \to a, S \to b$

$S \to a/b$ $\to$ LLR ✓ $\to$ RLG ✓ $\to$ Reg grammar ✓

$L = a^*$

$S \to aS/\epsilon \to RLG$

$S \to Sa/\epsilon \to LLG$

$S \to SS/\epsilon/a \to CFG$

$S \to SS/\epsilon/a/b$

$(a+b)^*$

$S \to SS/\epsilon/ab/ba$

$(ab+ba)^*$ ✓

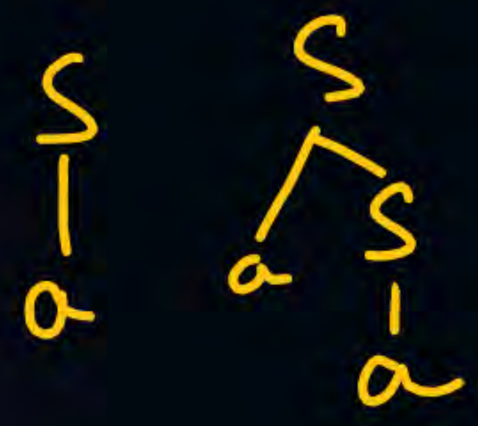
Some non regular grammars also generate Regular languages

$L = a^{+}$

$S \to aS / a \to RLG$

$S \to Sa / a \to LLG$

$S \to SS / a \to CFG$

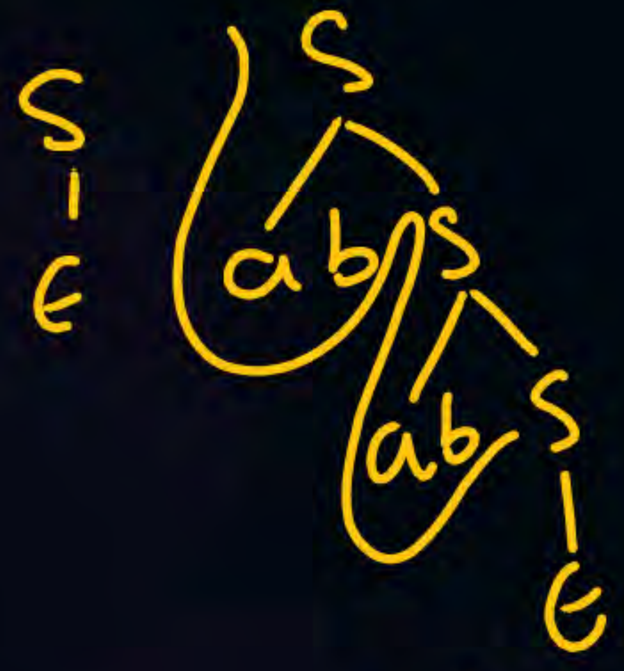


$(ab)^*$

$S \to abS / \epsilon$

$S \to Sab / \epsilon$

$S \to SS / \epsilon / ab$



abab

$(1101)^*$

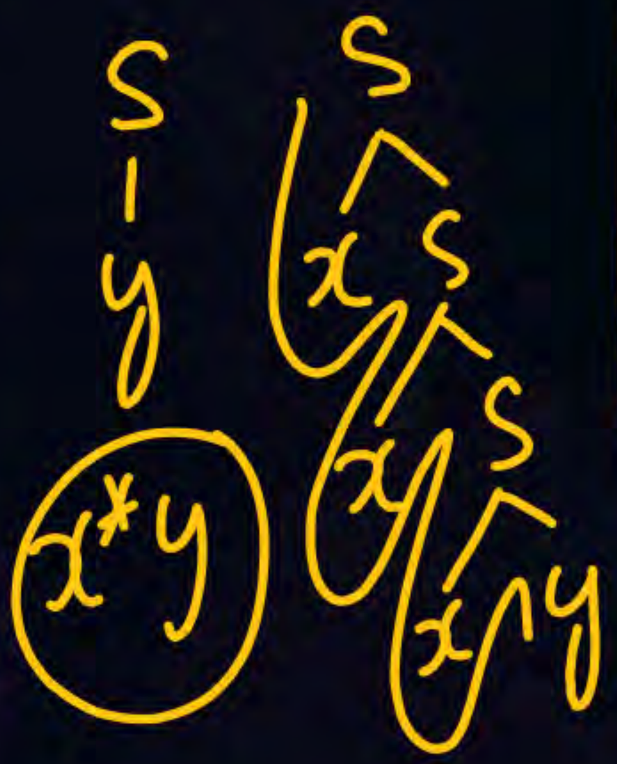
$S \to 1101S / \epsilon$

$S \to S1101 / \epsilon$

$S \to SS / \epsilon / 1101$

x^*y

$S \to xS / y$ → RLG



$S \to Ay$

$A \to xA / \epsilon$

→ CFG

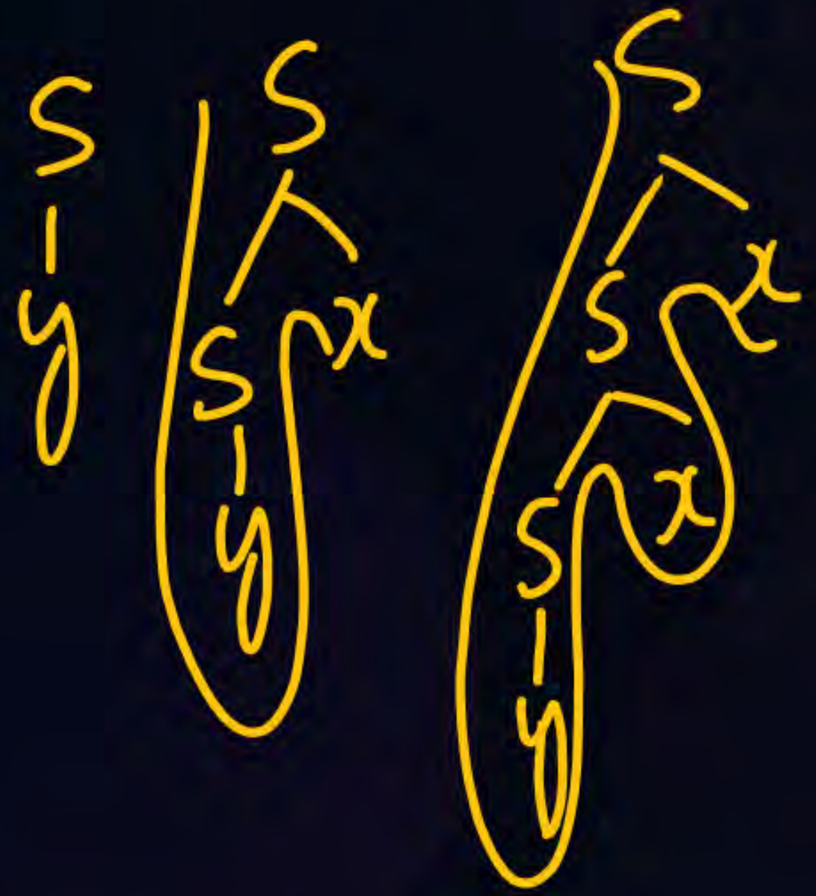
$S \to Ay$

$A \to Ax / \epsilon$

↓

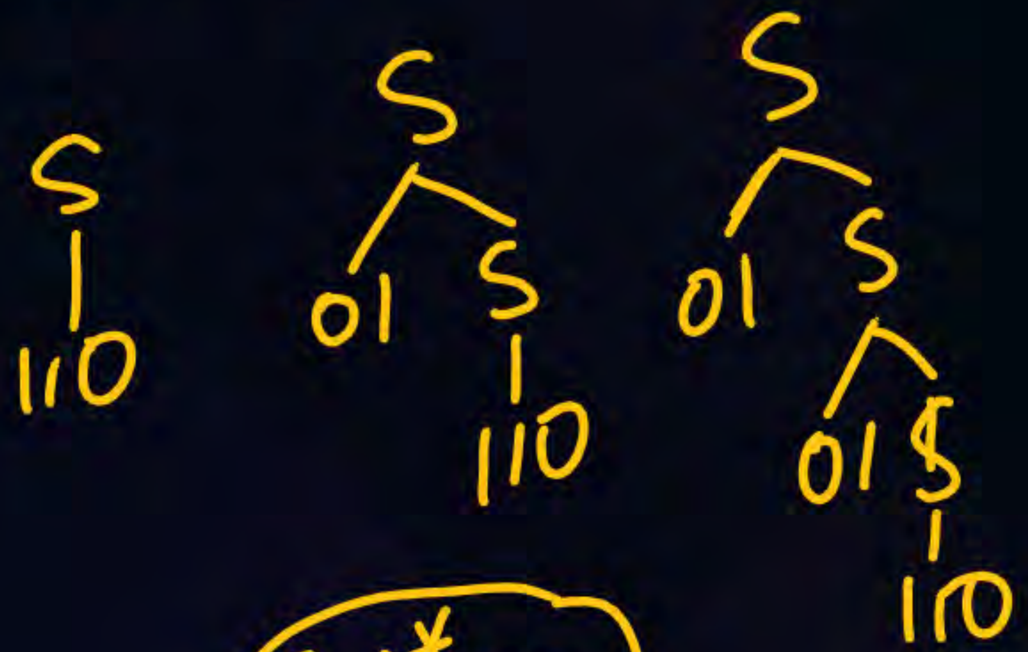
LLG

$S \rightarrow Sx / y$



yx^*

$S \rightarrow 01S / 110$



$(10)^* 110$

$L_1 \cup L_2$

$S \rightarrow S_1 / S_2$

L_1 L_2

$S_1 \rightarrow$

$S_2 \rightarrow$

$L_1 \cdot L_2$

$S \rightarrow S_1 \cdot S_2$

$a^* + b^*$

$$\left.\begin{aligned} S &\to S_1 / S_2 \\ S_1 &\to aS_1 / \epsilon \\ S_2 &\to bS_2 / \epsilon \end{aligned}\right\} \text{RLG}$$

$$\left.\begin{aligned} S &\to S_1 / S_2 \\ S_1 &\to S_1 a / \epsilon \\ S_2 &\to S_2 b / \epsilon \end{aligned}\right\} \text{LLG}$$

Gate

$$S \rightarrow A / B$$
$$A \rightarrow aaA / \epsilon$$
$$B \rightarrow Bb / \epsilon$$

$\left.\begin{matrix} \\ \\ \\ \end{matrix}\right\}$ CFG

a

$A \overset{*}{\Rightarrow} (aa)^*$ $\quad (aa)^* + b^*$

$B \overset{*}{\Rightarrow} b^*$

$S \rightarrow A / B$

$A \rightarrow aaA / \epsilon$

$B \rightarrow Baaaa / \epsilon$

$A \stackrel{*}{\Rightarrow} (aa)^*$

$B \stackrel{*}{\Rightarrow} (aaaa)^*$

0, 2, 4, 8, 6   0, 4, 8

$(aa)^* + (aaaa)^*$

$= (aa)^*$

$a^* b^*$

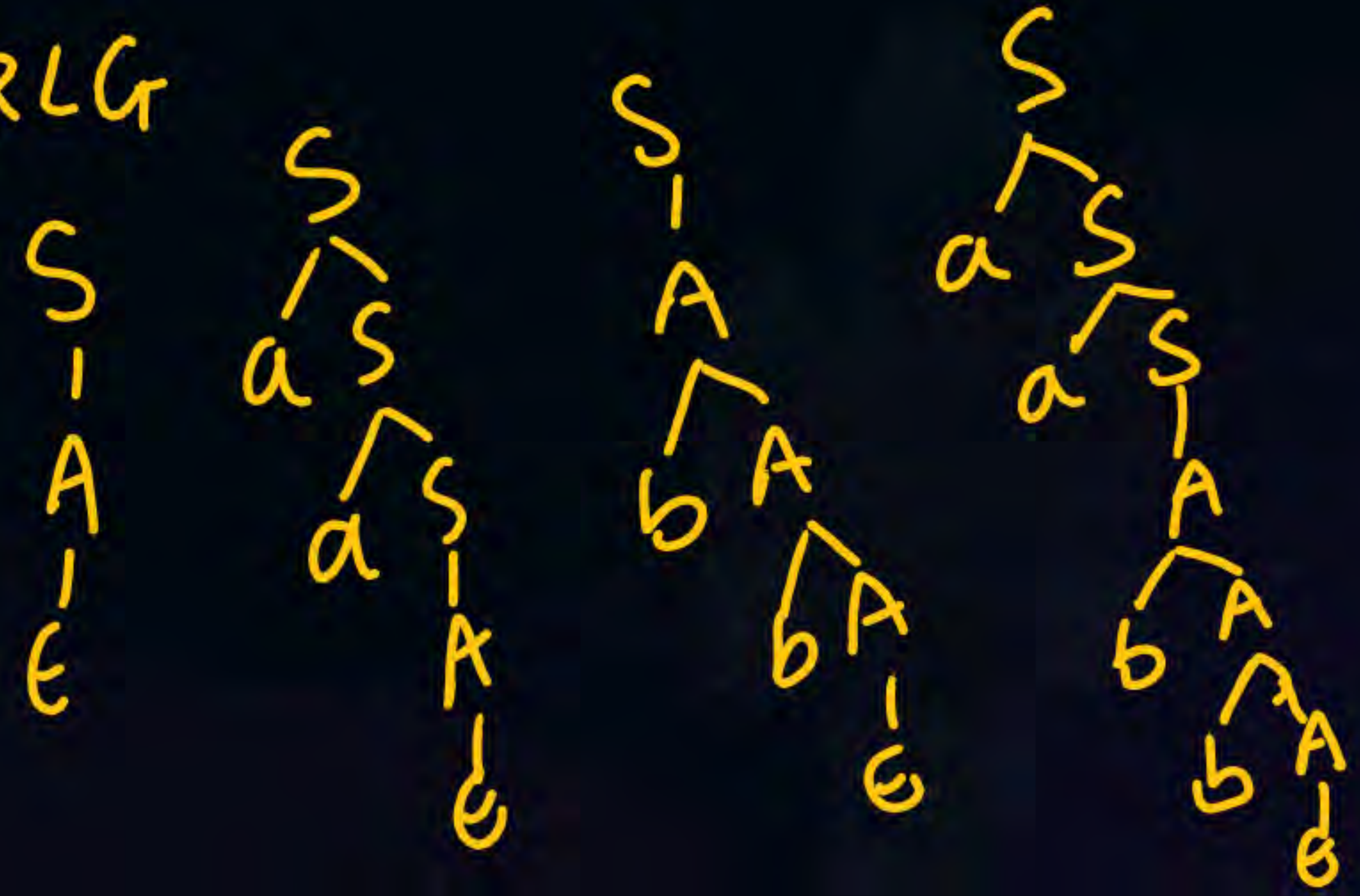
$S \rightarrow AB$
$A \rightarrow aA / \epsilon$
$B \rightarrow bB / \epsilon$

✓ CFG

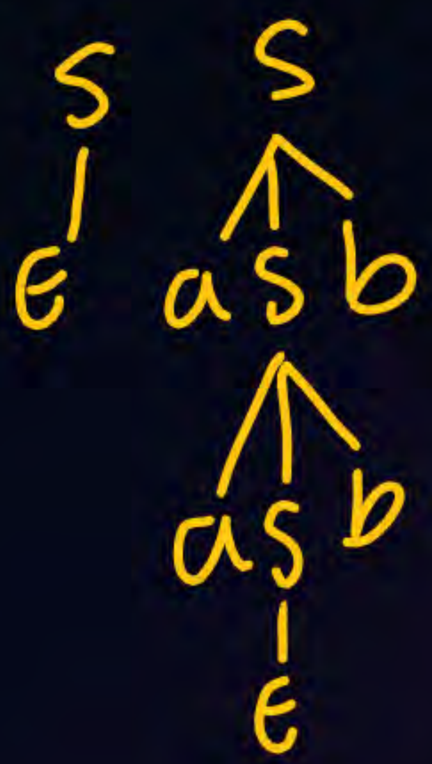
$S \rightarrow aS / A$
$A \rightarrow bA / \epsilon$

$a^* b^*$

RLG



$S \rightarrow aSb / \epsilon$



$a^n b^n \neq a^* b^*$

$a^* b^* \supseteq a^n b^n$

00
01
11
11

$(a+b)^*$

$S \to \underline{\underline{SS/\epsilon}} / \underline{\underline{a/b}}$ → CFG

$(a+b)^*$ ✓

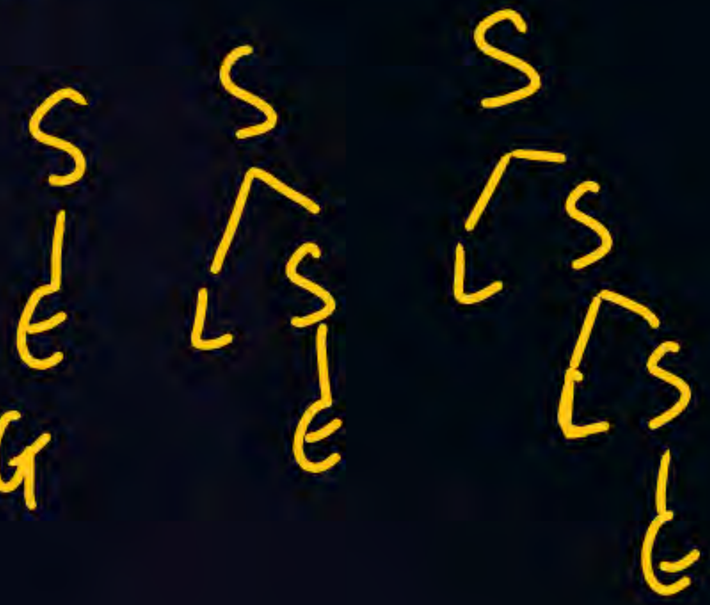
$S \to aS/bS/\epsilon \to$ RLG

$S \to Sa/Sb/\epsilon \to$ LLG

$L \rightarrow L^*$

$S \rightarrow SS/\epsilon/L \rightarrow$ CFG

$S \rightarrow LS/\epsilon \rightarrow$ CFG



$L \rightarrow$

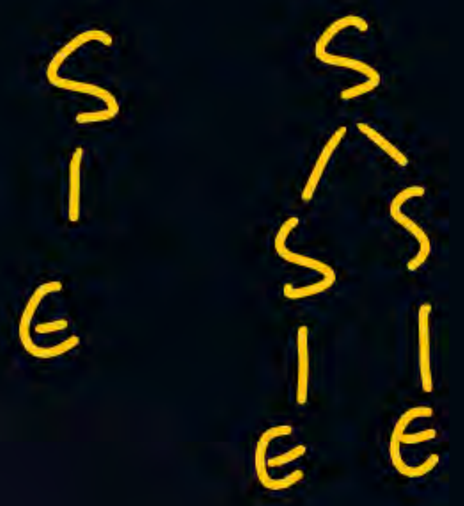
$(00+110)^*$

$S \rightarrow SS/\epsilon/00/110 \rightarrow$ CFG

$S \rightarrow 00S/110S/\epsilon \rightarrow$ RLG

$S \rightarrow S00/S110/\epsilon \rightarrow$ LLG

For any string if there are more than one parse tree, then the grammar is ambiguous



later

CFG

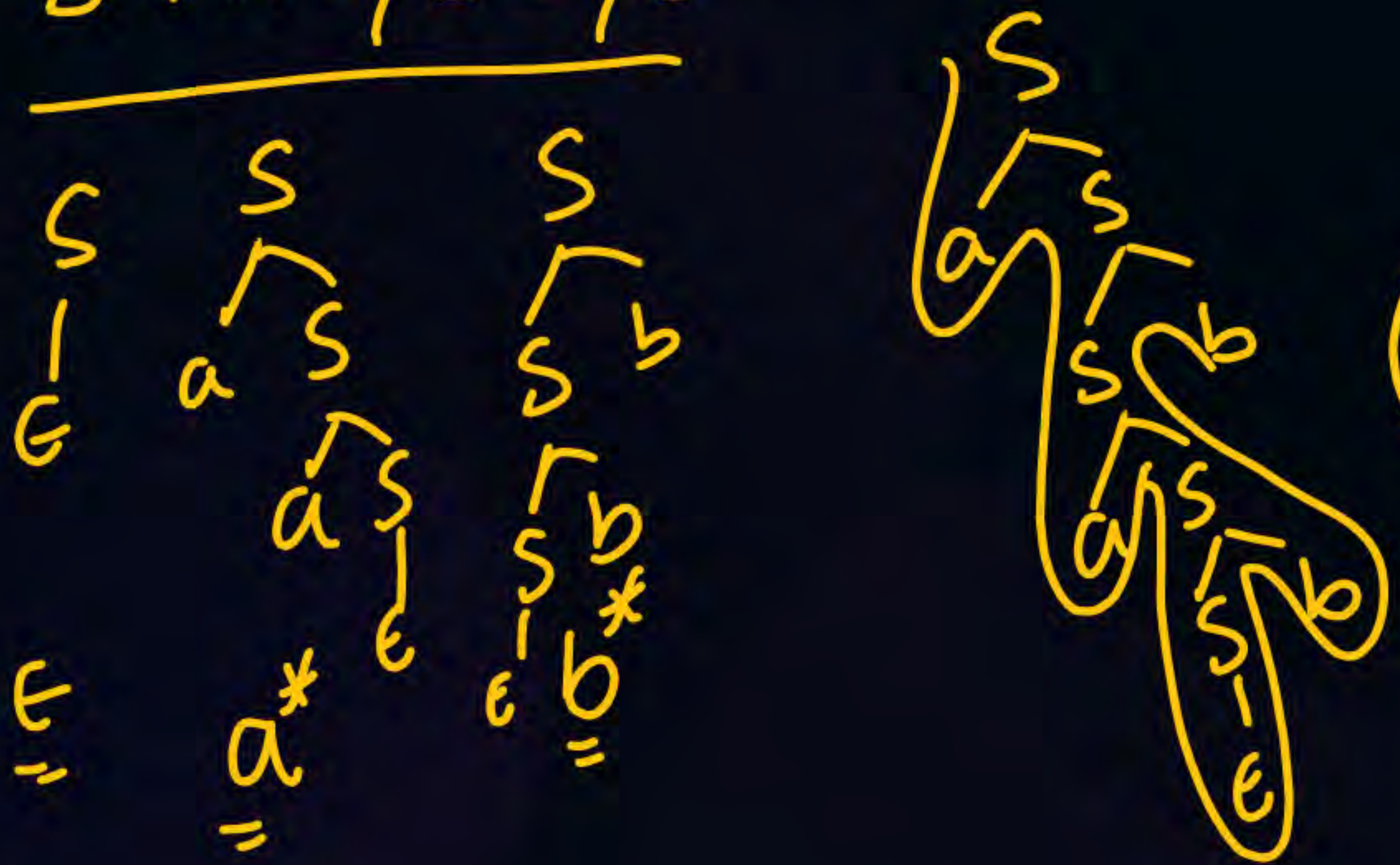
$(a+b)^+$

$S \to SS/a/b$

$S \to aS/bS/a/b \to RLG$

$S \to Sa/Sb/a/b \to LLG$

$S \rightarrow aS \,/\, Sb \,/\, \epsilon$ ✓



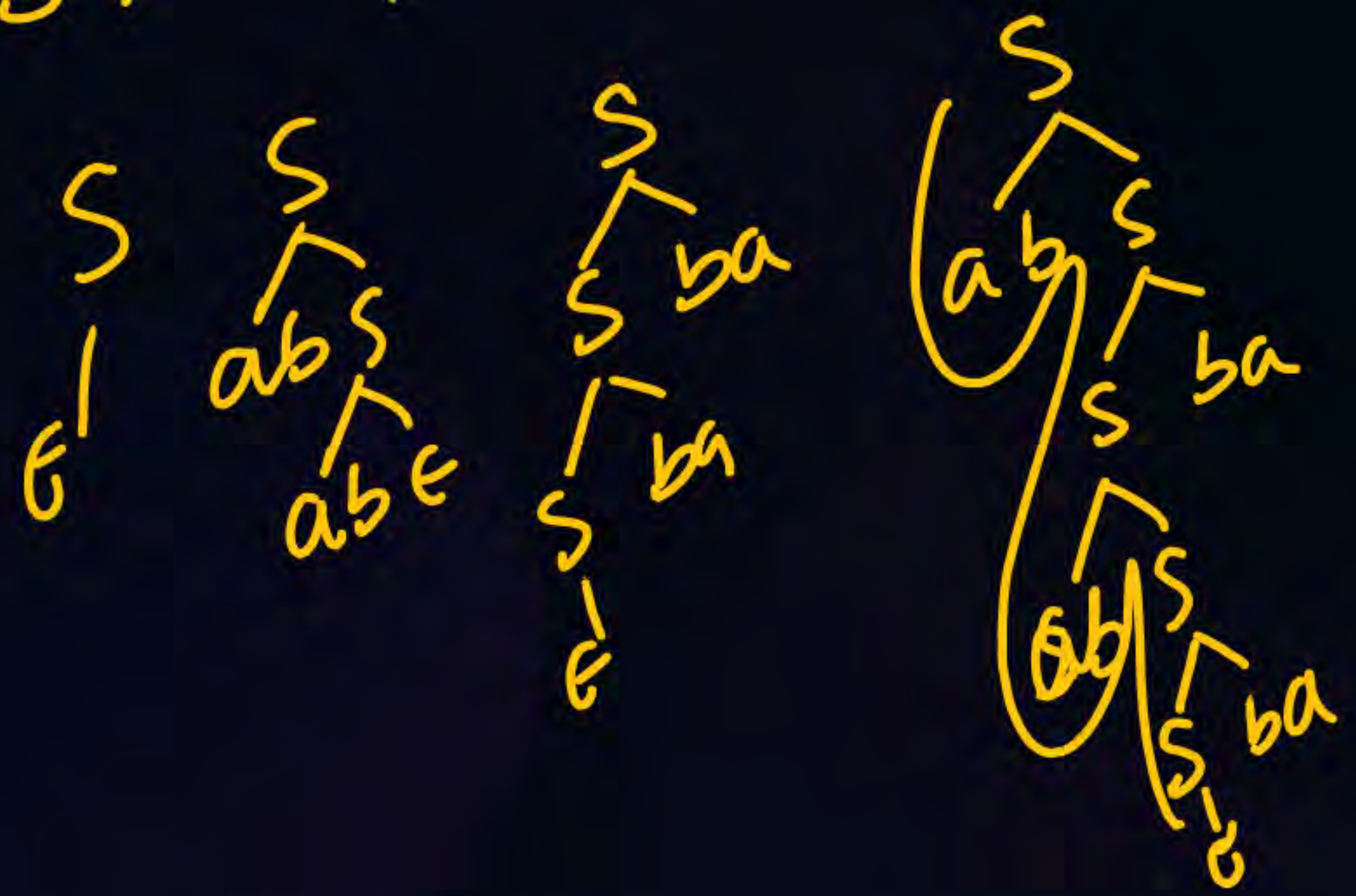
$a^* b^*$

$L \to FA \to Reg$

$L \to RE \to Reg$

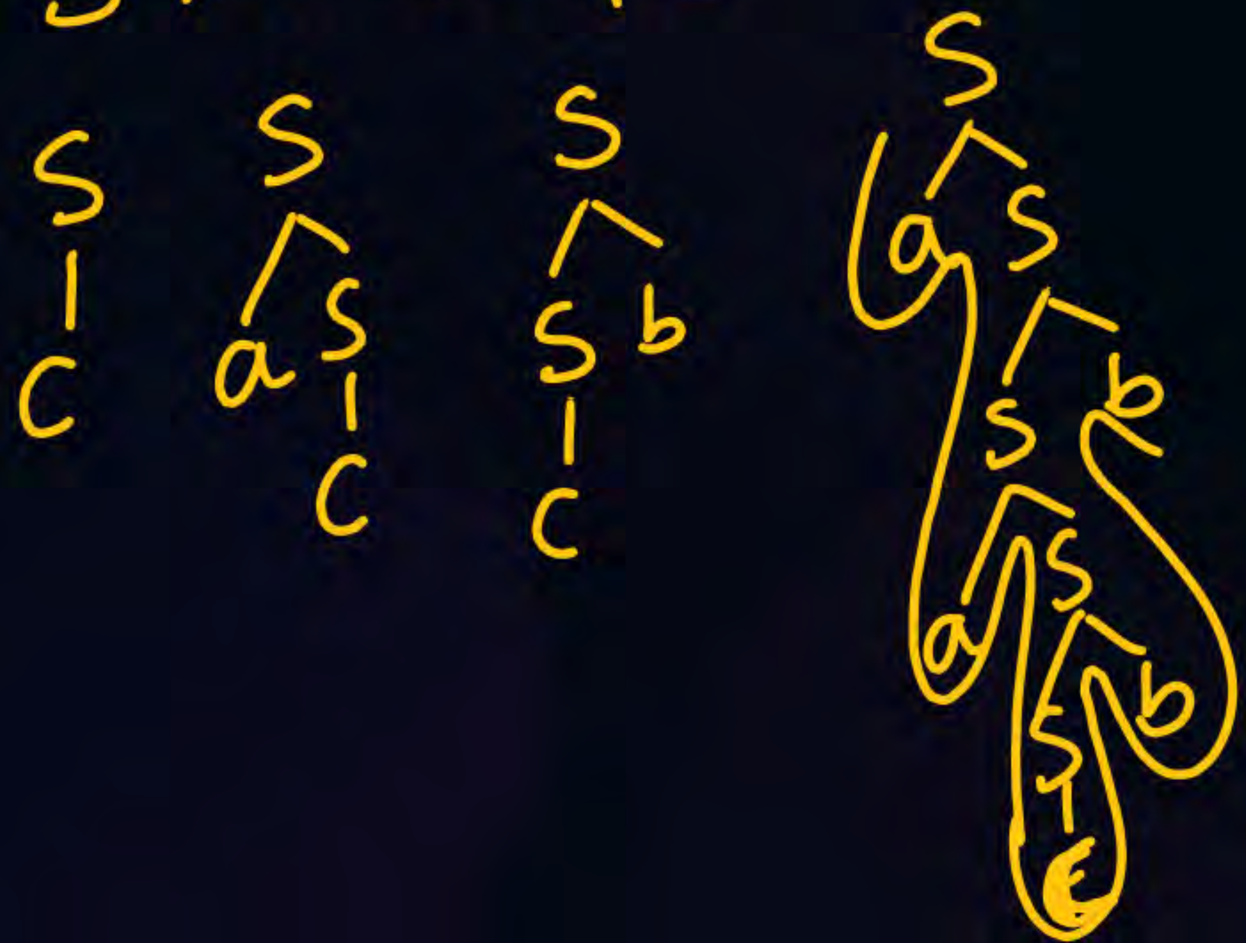
$L \to$ Regular gram $\to Reg$

$S \rightarrow abS / Sba / \epsilon$



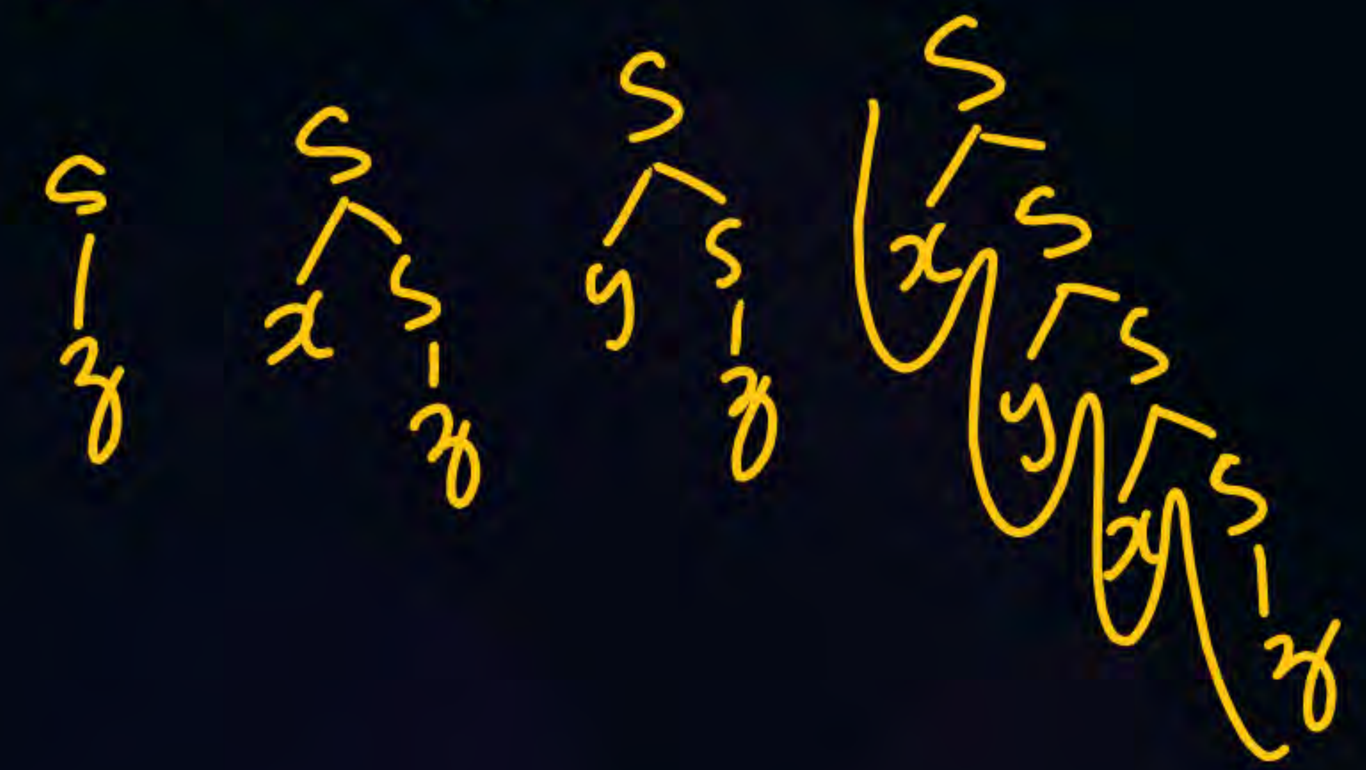
$(ab)^*(ba)^*$

$S \rightarrow aS / Sb / c$



$a^* c b^*$

$S \to xS / yS / z \to$ RLG

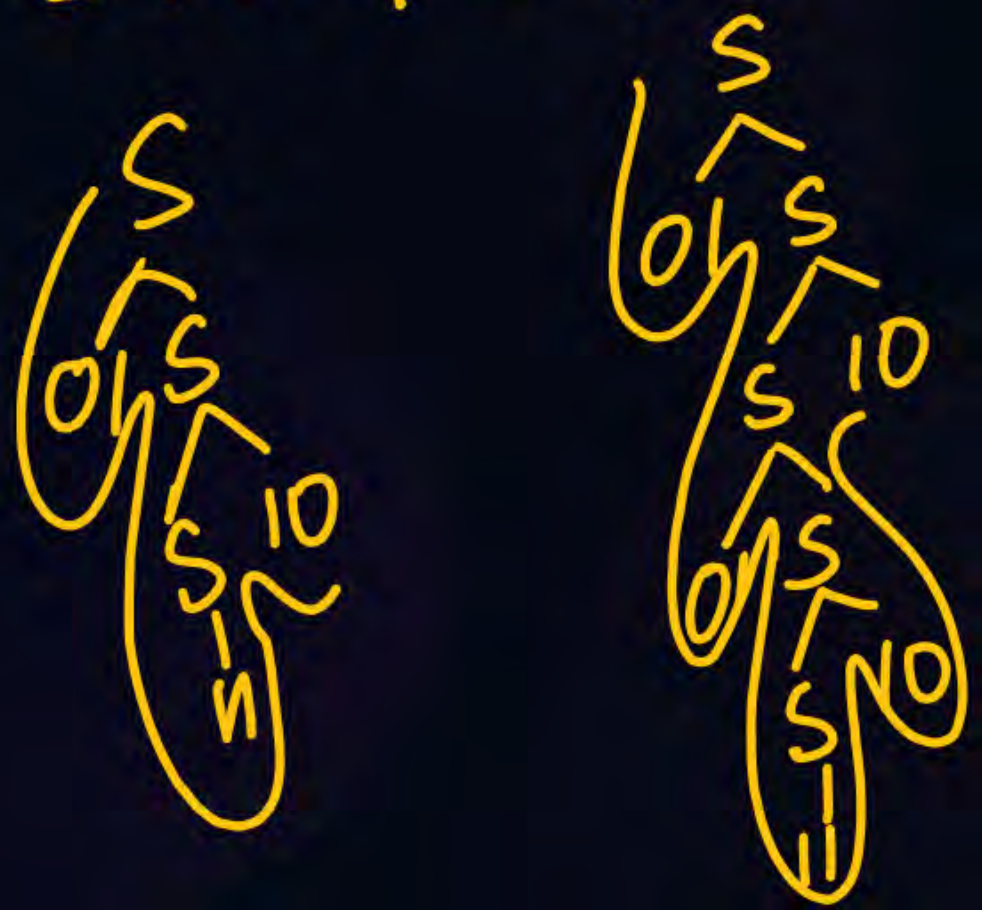


$(x+y)^* z$

$S \to Sx / Sy / z$

$z(x+y)^*$

$S \to 01S / S10 / 11$



$(01)^* \, 11 \, (10)^*$

$L = (a+b)(a+b)$

$S \to aa/ab/ba/bb$

$S \to AA$

$A \to a/b$

String length is atleast 2

$(a+b)(a+b)(a+b)^* \rightarrow$ Reg

$\rightarrow$ CFG

$S \rightarrow AAB$

$A \rightarrow a/b \rightarrow (a+b)$

$B \rightarrow aB/bB/\epsilon \rightarrow (a+b)^*$

FA $\rightarrow$ RLG

$\downarrow$

later

String length is atmost '2'

RE: $(a+b+\epsilon)(a+b+\epsilon)$

$S \rightarrow AA$

$A \rightarrow a/b/\epsilon$

Starting with 'a' and ending with 'b'

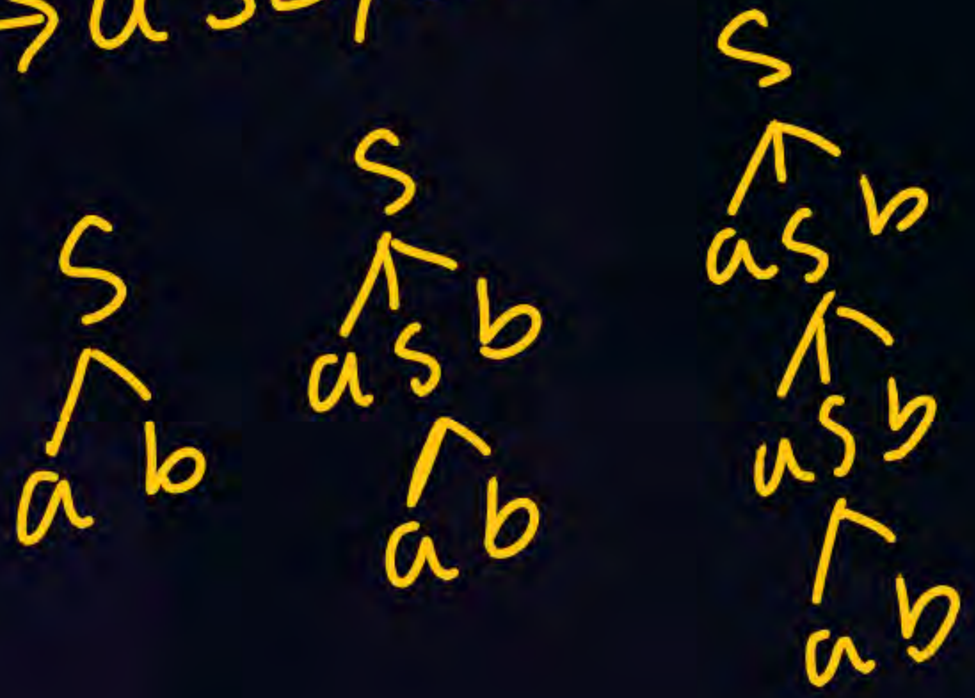
$$a\underline{\underline{(a+b)}}^* b$$

$$S \to aAb$$

$$A \to aA / bA / \epsilon$$

CFL $\leftarrow a^n b^n / n \geq 1 \rightarrow$ not reg

$S \rightarrow aSb / ab$



$a^n b^n / n \geq 1$

$\Sigma = \{a, b\}$ $L = ww^R + waw^R + wbw^R$

$w = abb$

$w^R = bba$

even length pal $\leftarrow ww^R = abbbba$ palidrome

$\checkmark$ a bbabb a

odd length pal $\left\{ \begin{array}{l} waw^R = abbabba \\ wbw^R = abbbbba \end{array} \right.$

$S \rightarrow aSa \,/\, bSb \,/\, a \,/\, b \,/\, \epsilon$

abba

$((a+b)(a+b))^*$ ✓

$$\left.\begin{aligned} S &\to BS/\epsilon \\ B &\to AA \\ A &\to a/b \end{aligned}\right\} CFG$$

$a^n b^m / n, m \geq 1$

$a^+ b^+$

$S \rightarrow AB$ ✓

$A \rightarrow aA / a$

$B \rightarrow bB / b$

$a^n b^n c^m / n, m \geq 1$

$S \rightarrow Ac$

$A \rightarrow aAb / ab$

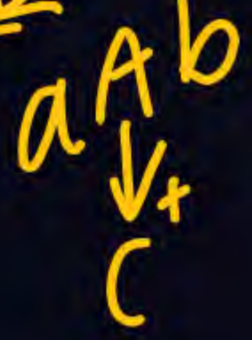
$C \rightarrow cC / c$

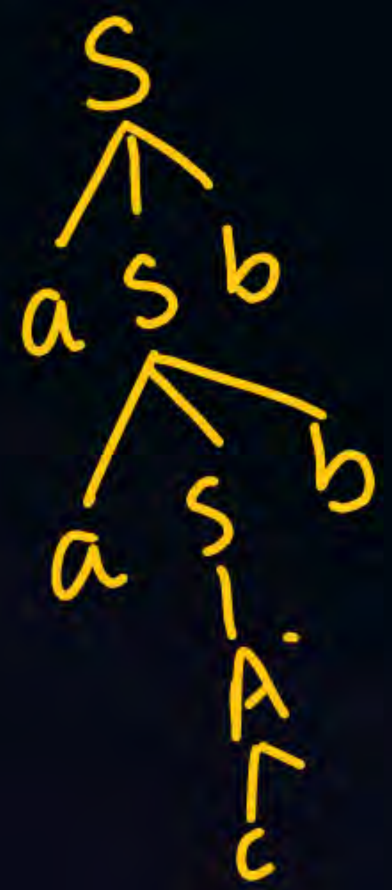
one minute

$a^n c^m b^n / n,m \geq 1$

$S \to aSb / aAb$

$A \to cA / c$





$a^n b^n c^m d^m / n, m \geq 1$ → Not Reg
→ CFL

$S \rightarrow AB$

$A \rightarrow aAb / ab$

$B \rightarrow cBd / cd$

$a^n b^n c^n / n \geq 1$

× Reg × CFL ✓CSL

↓

later

$a^n b^{2n} / n \geq 1$

$S \to aSbb \,/\, abb$

CFL × Reg

$\underline{a}bc\underline{d}$

$a^n b^m c^m d^n \,/\, n, m \geq 1$

$S \to aSd \,/\, \cancel{aAd}\ aAd$

$A \to bAc \,/\, bc$

S → CFG

A

bc

CSL → later

$a^n b^m c^n d^m \mid n, m \geq 1$

$\times$ Reg $\times$ CFL

$a^{m+n} b^m c^n / n, m \geq 1$

$a^m a^n b^m c^n$

$a^n a^m b^m c^n$

$S \to aSc / aAc$

$A \to aSb / ab$

$a^n b^{n+m} c^m / n, m \geq 1$

$\underline{a^n b^n}\ \underline{b^m c^m}$

$S \to AB$

$A \to aAb / ab$

$B \to bBc / bc$

$a^n b^m c^{m+n} / n \geq 1 \ m \geq 1$

CFL $a^n b^m c^m c^n / n \geq 1$

$$\text{CFG} \begin{cases} S \rightarrow aSc / aAc \\ A \rightarrow bAc / bc \end{cases}$$

RLG → FA

Rules

$A \to \epsilon \Rightarrow$ A is final state

$A \to a \Rightarrow A \xrightarrow{a} ◎$

$A \to ab \Rightarrow A \xrightarrow{a} \bigcirc \xrightarrow{b} ◎$

$A \to aab \Rightarrow A \xrightarrow{a} \bigcirc \xrightarrow{a} \bigcirc \xrightarrow{b} ◎$

$A \to aaaB \Rightarrow A \xrightarrow{a} \bigcirc \xrightarrow{a} \bigcirc \xrightarrow{a} B$

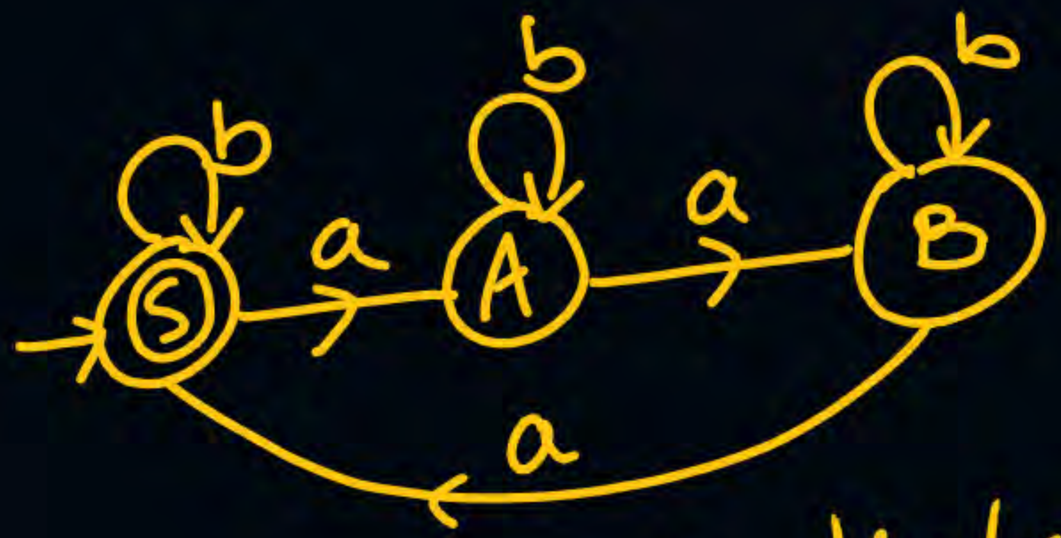
$A \to aab \Rightarrow A \xrightarrow{a} \bigcirc \xrightarrow{a} \bigcirc \xrightarrow{b} ◎$

$S \rightarrow aA / bS / \epsilon$

$A \rightarrow aB / bA$

$B \rightarrow aS / bB$

RLG $\rightarrow$ FA



No of a's div by 3.

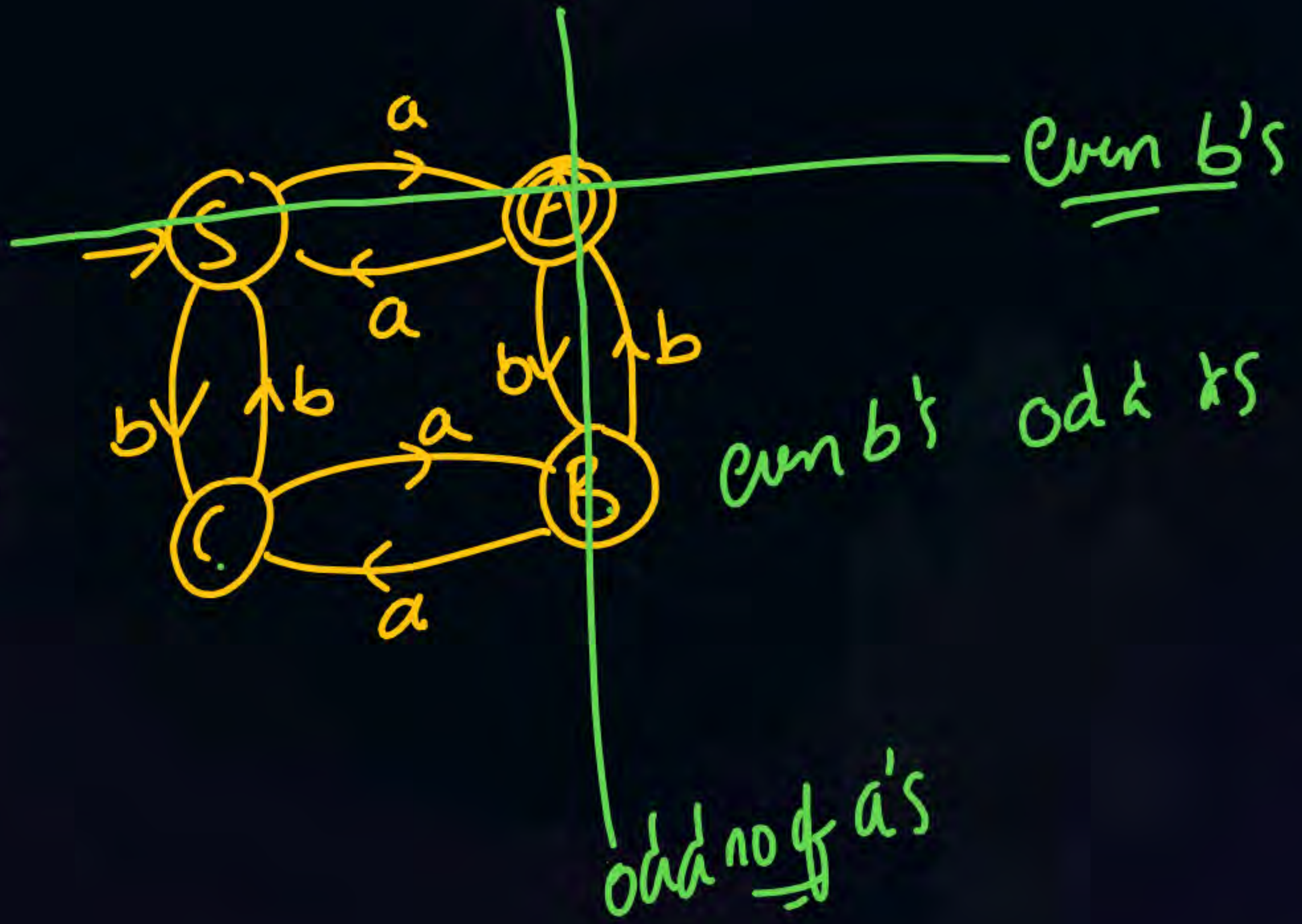
$S \rightarrow aA \,/\, bC$

$A \rightarrow aS \,/\, bB \,/\, \epsilon$

$B \rightarrow aC \,/\, bA$

$C \rightarrow aB \,/\, bS$

$A \rightarrow \epsilon$



RLG → FA

LLG (L) $\xrightarrow{\text{Reverse}}$ RLG (L^R) → FA (L^R) $\xrightarrow{\text{Reverse}}$ FA $(L^R)^R = L$

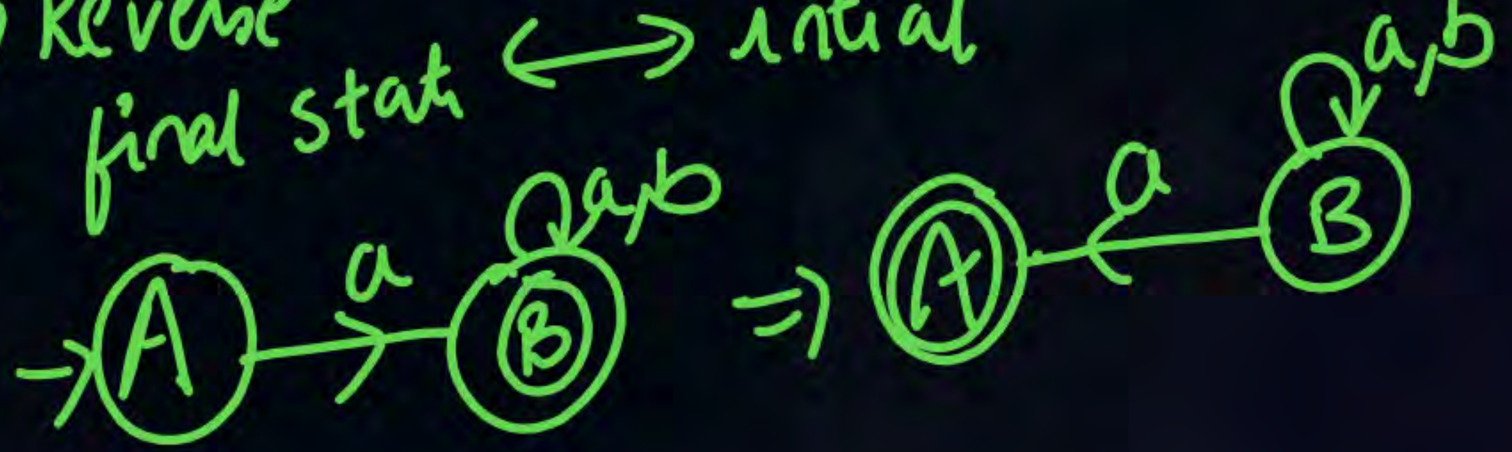
X exam

$A \to Aa/Ba/a$ $\quad$ $A \to aA/aB/a$

$B \to Bb/b$ $\quad$ $B \to bB/b$

FA → Reverse

make final state ⟷ intial



Algorithms in Reg languages:-

1) NFA → DFA
2) DFA → min DFA
~~3)~~ ϵ-NFA → NFA
4) mealy to moore
5) moore to mealy

→ $O(n^k)$ → polynomial

If for any problem there is an algorithm then it is called decidable

NFA → min NFA → exponential

→ $O(k^n)$

If algo is polynomial $\rightarrow$ Tractable and decidable

If algo is exponential $\rightarrow$ Intractable and decidable

DFA $\rightarrow$ NFA ✓

Every DFA is an NFA

NFA → DFA (Subset Construction method)

	a	b
→ q_0	q_0, q_2	—
* q_1	q_2	q_1
q_2	q_0	q_1

	a	b
→ $[q_0]$	$[q_0 q_2]$	D
D	D	D
$[q_0 q_2]$	$[q_0 q_2]$	$[q_1]$
* $[q_1]$	$[q_2]$	$[q_1]$
$[q_2]$	$[q_0]$	$[q_1]$

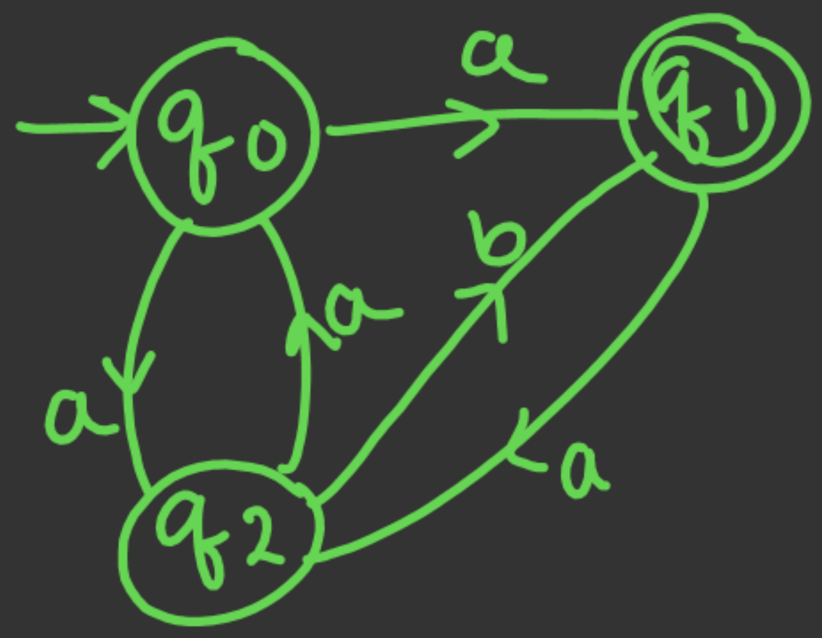
NFA

	a	b
$\rightarrow q_0$	q_1, q_2	—
$*q_1$	q_2	q_1
q_2	q_0	q_1

	a	b
$\rightarrow q_0$	$[q_1, q_2]$	D
D	D	D
$*[q_1, q_2]$	$[q_2\ q_0]$	$[q_1]$
$[q_2, q_0]$	$[q_0\ q_1, q_2]$	$[q_1]$
$*[q_1]$	$[q_2]$	$[q_1]$
$[q_2]$	$[q_0]$	$[q_1]$

	a	b
$*[q_0\ q_1, q_2]$	$[q_0\ q_1, q_2]$	$[q_1]$

Whenever we convert NFA $\rightarrow$ DFA using Subset Construction method, we may or may not get min <u>DFA</u>.

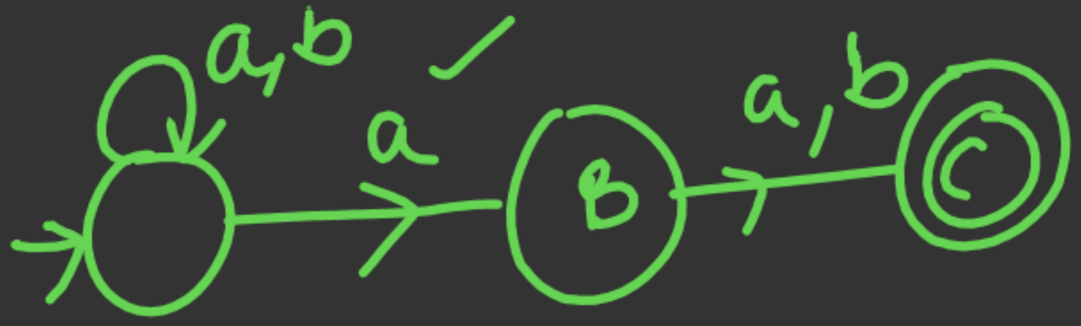


DFA → 7 States

FS → 3 states

Second Symbol from RHS is a

$$(a+b)^* \underline{\underline{a}}(a+b)$$



NFA – 3 states

THANK - YOU